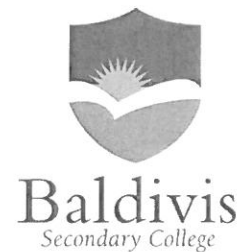


Year 10
Pathway 2

Test 2



Name: Solutions

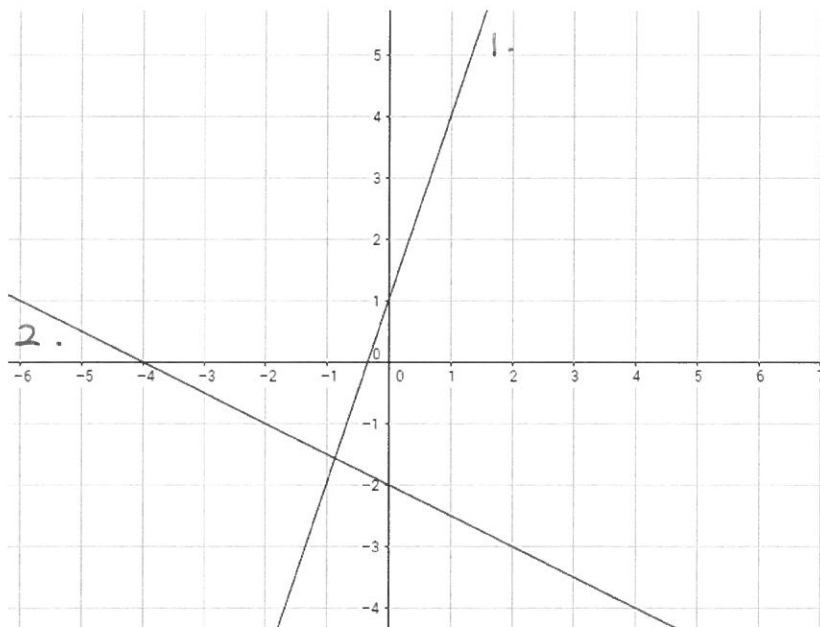
Total Marks: 28 marks
Total Time: 40 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources allowed: 1 page of notes A4, 1 side. Calculator

1. Identify the gradient and y intercept of the two lines below:

(6 marks)



line	Gradient	Y intercept
Line 1	3 ✓	1 ✓
Line 2	$-\frac{1}{2}$ ✓	-2 ✓

Hence state the equations of the lines:

Line 1: $y = 3x + 1$ ✓

Line 2: $y = -\frac{1}{2}x - 2$ ✓

2. Complete the table of values for the following quadratic relationship

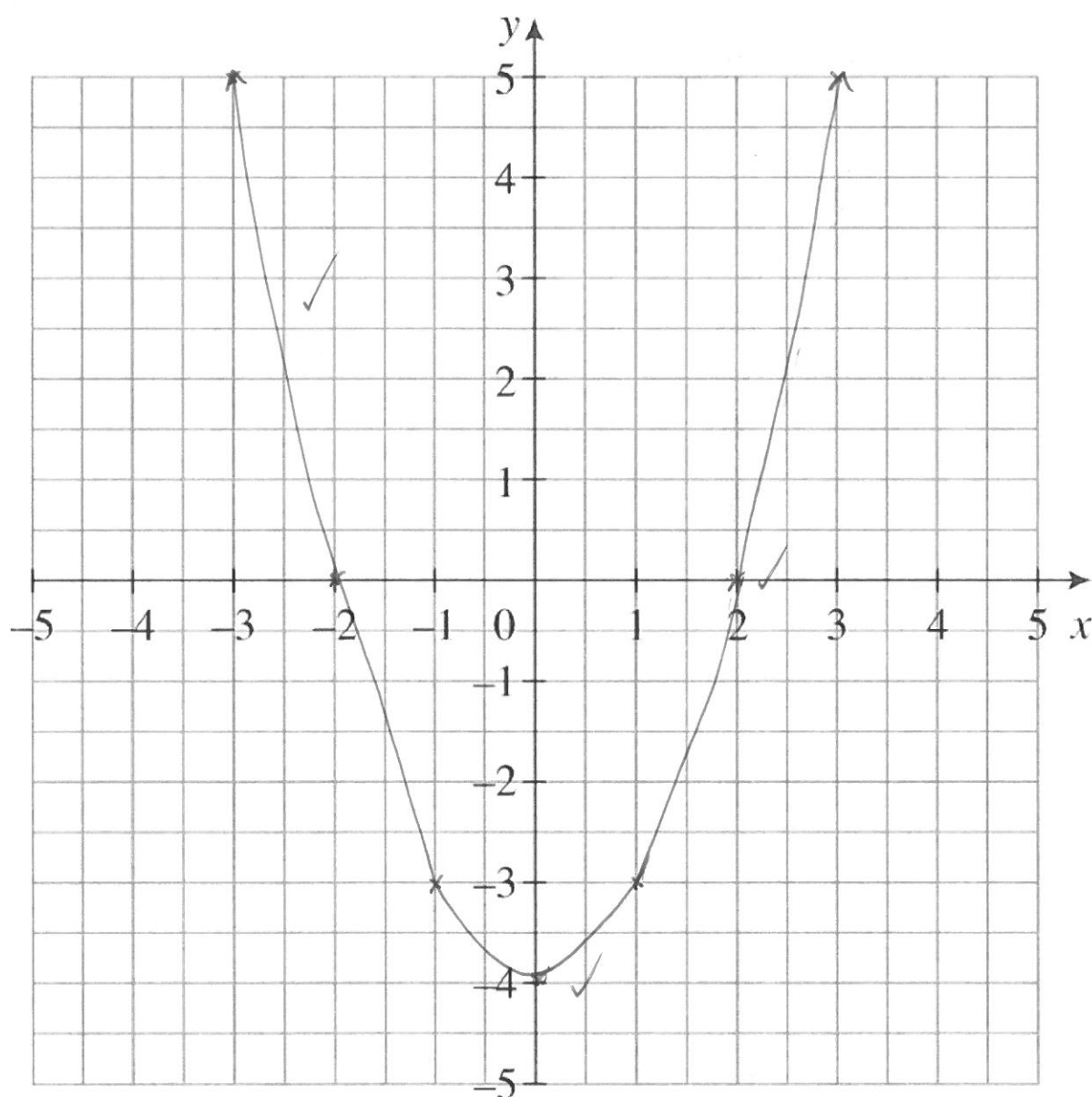
(7 marks)

x	-3	-2	-1	0	1	2	3
$y = x^2 - 4$	5	0	-3	-4	-3	0	5

✓ ✓ ✓ ✓ ✓ ✓ ✓

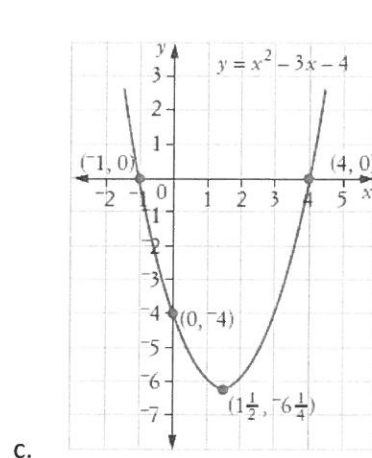
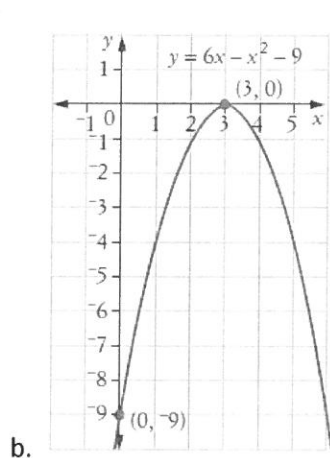
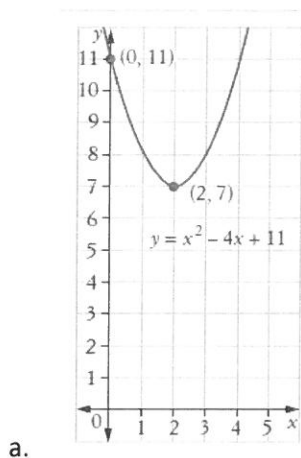
3. Plot the above relationship on the axes given below

(3 marks)



4. Fill in the table for the graphs given below:

(12 marks)



Line	Equation	Max/Min	Turning Point	Y intercept	Axis of Symmetry
a	$y = x^2 - 4x + 11$	min ✓	(2, 7) ✓	11 ✓	$x = 2$ ✓
b	$y = 6x - x^2 - 9$	max ✓	(3, 0) ✓	-9 ✓	$x = 3$ ✓
c	$y = x^2 - 3x - 4$	min ✓	$(1\frac{1}{2}, -6\frac{1}{4})$ ✓	-4 ✓	$x = 1\frac{1}{2}$ ✓

END OF TEST

